

1| 2| 3| 4| 5| 6| 59| 60| 61|
62|

Arinze Elenasulu

63|

Python Backend Engineer | AI & Automation

64|
65| London, UK · elenasuluarinze@gmail.com
66| [linkedin.com/in/arinze-elenasulu](https://www.linkedin.com/in/arinze-elenasulu) · 67| github.com/haxsysgit 68|
69|
70| 71|
72|
Professional Summary
73|
74|

I am a recent BSc Information Technology graduate from Middlesex University London, and I have been writing Python since 2020. During my studies, I built the backend for a pharmacy operations platform from scratch, designed governance tooling for AI coding agents published on PyPI, and completed a two-year Advanced Diploma in Software Engineering covering Java, Spring, C++, and full-stack web development. I also had a software engineering role at Vigilis where I owned the backend for a production system. I learn fast, I write clean code, and I am looking for a graduate or junior engineering role where I can contribute from day one and keep growing.

75|
76|
77| 78|
79|
Core Skills
80|

Backend Engineering: Python (primary, since 2020), FastAPI, SQLAlchemy, Alembic, REST API design, PostgreSQL, schema design
AI & Automation: LLM integration, MCP tooling, RAG pipelines, AI governance systems, agentic workflows
Infrastructure & Tools: Docker, Linux, Git, structured logging, defensive backend design
Testing: Advanced pytest (fixtures, parametrisation, integration coverage), API test automation
Languages & Stacks: Java, Spring Framework, C++, C, JavaScript/TypeScript, React, MERN stack, MEAN stack, Flutter/Dart (from formal training)

81|
82| 83|
84|
Experience
85|

Vigilis, Software Engineer
August 2024 – February 2026 · Lagos, Nigeria
Built and maintained the backend for a pharmacy operations platform handling sales, inventory, payments, and reporting across multiple locations. System used day-to-day by pharmacy staff.

- Designed the full data model and API layer (FastAPI, PostgreSQL, SQLAlchemy) with role-based access, JWT auth, and Alembic migrations that kept the database evolving safely as features shipped.

- Sat with pharmacy staff directly to understand their manual processes. Turned paper workflows and mental checklists into software that cut daily admin time and reduced stock discrepancies.
- Built the invoice lifecycle: create, add line items, finalise with automatic stock deduction, cancel with stock reversal. Every state transition validated and audited.
- Added structured logging, input validation, and business rule enforcement that caught errors at the API boundary before they could corrupt data downstream.
- Owned the backend end to end: feature planning, database design, deployment, monitoring, and production bug fixes. No handoff.

Aptech Computer Education, Software Engineer Intern

September 2022 – August 2024 · Lagos, Nigeria

Completed the Advanced Diploma in Software Engineering (ADSE Java track), a structured two-year programme combining classroom training with hands-on project work.

- Covered the full software engineering stack across four semesters: Linux system administration, Python, C++, C, Java, Spring Framework, Enterprise Java Beans (EJB), Flutter/Dart for mobile, and both MERN and MEAN stacks for full-stack web development.
- Built multiple projects across different technology stacks as part of the programme curriculum, gaining practical exposure to how different languages and frameworks solve similar problems.
- Developed strong fundamentals in object-oriented programming, database design, and software architecture that carried into every role since.
- Python became my primary language during this period and has remained so. I have been studying and building with it continuously since 2020.

Bucca Hut, AI and Backend Engineer (Contract, part-time)

February 2025 – May 2025

Built a data analysis tool for a food business, connecting LLM outputs to real operational decisions. Designed the backend API bridging AI processing with the existing workflows.

86|

87| 88|

89|

Selected Projects

90|

FRAME, Typed Project Memory Format for AI Agents

Python, Markdown/YAML, Multi-Agent

github.com/haxsysgit/FRAME

Designed a portable standard that gives AI coding agents structured project context across sessions.

- Five-part memory model (Facts, Rules, Acts, Map, Expect) capturing project truth, agent rules, session history, module relationships, and expected outcomes.
- Reduces repeated prompting, context rot, and full-repo rescans across agent sessions.
- Designed as a portable standard: any AI agent reads the same project representation.
- Published docs, learn materials, and frame-python implementation track on GitHub.
- Separation of concerns: FRAME is the contract, implementations earn alignment with it.

Haxaml, AI Agent Governance & Project Memory

Python, MCP, FRAME

github.com/haxsysgit/haxaml · pypi.org/project/haxaml

Built and published a governance runtime that enforces project rules for AI coding agents.

- Governance runtime enforcing FRAME correctness contract, blocks AI-generated code until mechanical checks pass.
- Five-stage pipeline: Admission → Scoped Context → Agent Work → Verification → Recording.
- MCP server so AI coding tools can read and write to FRAME files during a session.
- Local dashboard for inspecting agent session history and project state.

- Published on PyPI with documentation, setup flow, and adoption guides.

Pharmax, Pharmacy Management Platform

Python, FastAPI, PostgreSQL, SQLAlchemy, Alembic

Built the backend for a pharmacy operations platform handling product inventory, sales, invoicing, and reporting.

- Products CRUD with multi-unit pricing (tablets, packs, strips).
- Full invoice lifecycle: draft to finalise (auto stock deduction) to cancel (stock reversal).
- Stock adjustments with audit trail and reorder-level tracking.
- Role-based access with JWT authentication.
- Comprehensive pytest suite covering endpoints, business logic, and edge cases. Sat with pharmacy staff to understand manual processes and turned them into software.

HaxJobs, Automated Job Search & Application Platform

Python, Playwright, React, TypeScript, Hermes Agent

github.com/haxsysgit/haxjobs

Built an end-to-end automated job discovery and application platform powered by AI agent infrastructure.

- Multi-platform job discovery across 95+ companies (Lever, Ashby, Greenhouse APIs).
- Hermes-powered job evaluation pipeline scoring fit to user profile.
- 5-level automated application pack generation with CV, cover letter, Q&A.
- Web dashboard for pipeline monitoring with real-time job status views.
- Zero-token discovery via system-level cron scheduling, independent of LLM.

91|

92| 93|

94|

Education

95|

BSc Information Technology , Middlesex University London (Expected June 2026)

96|

Advanced Diploma in Software Engineering (ADSE Java) , Aptech Computer Education, Lagos (September 2022 – August 2024)

97|

98| 99| 100|